

OSF Amendment 02 — Latin-speaking provinces as the primary hypothesis-testing frame

title: OSF Amendment 02 — Latin-speaking provinces as the primary hypothesis-testing frame

amendment-number: 02

status: LODGED 2026-06-06 (OSF; Latin-frame confirmatory gate cleared)

date-drafted: 2026-06-06

scope: Reframe the primary cross-sectional hypothesis-testing frame (H3a, H3b, H3c, SR1) from empire-wide to Latin-speaking provinces, on a dataset-coverage rationale; retain empire-wide results as secondary/context; promote the province→language map to a first-class tracked artefact; reconcile the realised 39-province frame against the lodged prereg’s 41-province figure.

preregistration: <https://osf.io/uycs6/> (lodged 2026-05-20; embargoed)

lodged-version: git tag osf-lodgement-2026-05-20 (<https://github.com/saross/inscriptions/tree/osf-lodgement-2026-05-20>)

filed-under: preregistration §7 / contingency clause (preregistration-draft.md line 423): substantive methodology changes after lodgement are filed as an OSF amendment before implementation.

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gate: CLEARED 2026-06-06 — the amendment is lodged; Latin-frame confirmatory claims (H3a/H3b/H3c/SR1 on the Latin-speaking-province subset) may now leave the repository as confirmatory. The empire-wide H3a result was within the original lodged prereg text and was never gated by this amendment.

relationship: Independent of Amendment 01 (two-measure framework, lodged 2026-06-04) and of the forthcoming convention-model amendment (Decision 38 empirical convention basis, which gates the H2.1 temporal-mixture fit). All three are separable; this one concerns only the cross-sectional hypothesis-testing frame, which uses date-window-filtered counts, not mixture output.

OSF Amendment 02 — Latin-speaking provinces as the primary hypothesis-testing frame

Plain-language summary

This project measures how much of the city-to-city variation in Roman inscription production is explained by city population, and whether that pattern reproduces a published finding (Hanson 2021). The dataset we use, LIRE (“Latin Inscriptions of the Roman Empire”), is what its name says: a corpus of *Latin* inscriptions. In the Latin-speaking western provinces it is close to complete; in the Greek-speaking east, Latin inscriptions are a minority of what was actually produced, so LIRE captures only a non-representative slice of those provinces’ epigraphy (and dedicated Greek corpora are still being built). An empire-wide analysis therefore mixes well-covered Latin provinces with poorly-covered Greek ones — a coverage artefact, not a fact about the ancient world.

This amendment makes one change: it promotes the **Latin-speaking provinces** to the primary frame for the hypothesis-testing analyses, and demotes the empire-wide analysis to reported-but-secondary context (with the coverage caveat stated). It does not change a single model, decision rule, or threshold. The empire-wide result is kept and reported, not hidden. Two supporting points: (1) the rationale is about data coverage, which is true regardless of what any result shows; and (2) the lodged preregistration’s own “~815 cities” figure already **was** the Latin-province filter — the Latin frame was latent in the prereg’s own number, and the broader “all cities” reading is what introduced the coverage confound. The amendment also reconciles a bookkeeping discrepancy (the

prereg quotes 41 Latin provinces; the realised analysis frame has 39) and shows it changes no result. The technical statement follows.

A1. Identification and trigger

This is the second amendment to the project’s preregistration (Open Science Framework, OSF, record osf.io/uycs6/, lodged 2026-05-20, currently embargoed). It is filed under the preregistration’s own contingency rule (`preregistration-draft.md` line 423): “If substantive methodology changes are required after lodgement ... an OSF amendment is filed before implementation.”

It is a **binding gate** for the Latin frame only: Latin-speaking-province confirmatory claims (H3a, H3b, H3c, SR1) do not leave the project repository as confirmatory until this amendment is lodged. The empire-wide H3a result is within the original lodged preregistration’s text (“all cities”) and is **not** gated by this amendment — it is retained and reported as secondary/context.

Trigger. The H3a confirmatory run (a blind fit, the analyst writing it kept separate from the talk-preliminary thread) produced the within-province population-attributable variance fraction `f_within` on two frames: empire-wide and a Latin-speaking-province subset. Reviewing the two, two things became clear: (i) the empire-wide frame carries a structural **dataset-coverage confound** (LIRE under-represents Greek-province epigraphy), which makes the Latin-province frame the methodologically defensible primary; and (ii) the preregistration’s own city-count figure (“~815”) was, on inspection, already the Latin-province filter — so the reframe corrects an under-specification in the lodged document rather than introducing a new analysis. Decision 36 (`planning/decision-log.md`) records the deliberation.

A2. Summary of the change

The amendment makes **one** connected change, with one bookkeeping reconciliation.

(1) **Latin-speaking provinces become the primary hypothesis-testing frame.** For the cross-sectional confirmatory analyses — **H3a** (the primary confirmatory result), **H3b**, **H3c(i)/(ii)**, and **SR1** — the primary unit set is the Latin-speaking provinces of LIRE v3.0 (Rome excluded), defined by the project’s province→language classification. Empire-wide results are reported as **secondary/context**, with the coverage caveat stated. The unit is **Latin-speaking provinces, not Latin-only inscriptions** — the rationale is provincial *coverage* (the dataset is approximately complete for Latin provinces, incomplete for Greek ones), so the province-level frame is the one that matches the rationale; a language-of-inscription filter was considered and rejected (bilingual/mixed contexts, language-classification noise, and it abandons the coverage logic that is the actual motivation).

(2) **Frame-definition reconciliation (41 → 39).** The lodged prerog (§2) defines the Latin/“Western-Empire” subset as **41 LIRE provinces** (Rome excluded), from the 2024 exploratory notebook’s `province_language_map`. The realised primary frame is **39 provinces / 817 cities**. §A5.3 reconciles the difference from evidence and shows it moves no result.

This is a change of **analysis frame**, not of model, estimand, or decision rule. Every model specification, prior, confirmatory rule, and threshold in the lodged protocol is unchanged (§A6).

A3. Rationale

The dataset is a Latin corpus. LIRE — “Latin Inscriptions of the Roman Empire” (Kaše, Heřmáňková & Sobotková, Zenodo DOI 10.5281/zenodo.8431452) — is drawn from the larger LIST (“Latin Inscriptions through Space and Time”). In Greek-speaking provinces, Latin inscriptions are a distinct minority of epigraphic production, so LIRE captures only a non-representative Latin slice of those provinces; Greek epigraphic coverage in the corpus is inconsistent, and dedicated Greek datasets are still in production. Empire-wide hypothesis-testing therefore pools well-covered Latin provinces with poorly-covered Greek ones — a confound for any claim about epigraphic *production* (as opposed to *Latin*-epigraphic production). The Latin-speaking-province frame removes the confound by restricting to the part of the empire the dataset actually covers well.

The prerog’s own number already was the Latin filter. The lodged preregistration twice anchors the cross-sectional sample at “~815 cities” (§“the cross-city H3a ... uses *all* ~815 cities with Hanson population estimates after Rome-exclusion”). The realised empire-wide (“all cities”) frame is **1,044** cities; the realised **Latin**-province frame is **817**. The prerog’s “~815” is the Latin filter, not the empire-wide count — the Latin frame was latent in the

lodged document’s own figure. The over-broad “all cities” reading is what introduced the confound. As recorded in Decision 36: “it should have been specified in the original prereg, I just missed it.” The reframe therefore *corrects an under-specification* rather than swapping in a new, post-hoc choice.

This is coverage-driven, not result-driven (see §A4). The coverage argument is true independent of any result; the empire-wide result is retained and reported; no decision rule is tuned to any observed Latin-frame number.

A4. Relationship to already-observed results (transparency)

Full disclosure, because the reframe follows a run whose results are already seen: on the Latin-province frame the within-province population effect is **stronger** than empire-wide (f_{within} 0.480 vs 0.299; see §A5.4). A reviewer is entitled to ask whether the frame was chosen because it gave the larger number. Four points bear on that, and we state them rather than hide them:

1. **The rationale is independent of the result.** The coverage confound (LIRE under-covers Greek epigraphy) is a property of the *dataset*, established by what LIRE is, not by any f_{within} value. It would be the reason to prefer the Latin frame whether the Latin f_{within} came out larger, smaller, or equal.
2. **The prereg’s own “~815” was the Latin filter (§A3)** — the Latin frame is the lodged document’s latent sample, not a new construct introduced to chase a result.
3. **The empire-wide result is retained and reported**, as secondary/context with the coverage caveat (§A5.5). Nothing is suppressed; both frames appear in the paper.
4. **No decision rule changes.** The three-way f_{within} rule, the H3c rules, and SR1’s comparator are identical to the lodged protocol (§A6). The Latin verdict is read off the same machinery, not a frame-tuned one.

The honest summary: the frame change is a methodological correction with an independent justification; it happens to *raise* the headline effect, and we report both frames so the reader can see exactly that.

A5. Pre-specifications

A5.1 Primary frame definition

The primary hypothesis-testing frame is the set of LIRE v3.0 **Latin-speaking provinces**, Rome excluded, defined by the project’s province→language classification, operationalised as the committed artefact `runs/2026-06-04-h3a-confirmatory/data/province-language-map.csv` (the externalised, LIRE-province-field-aligned form of the 2024 notebook’s `province_language_map`, `archive/2026-04-22-inscriptions-spa.ipynb` cell 54). The map is **promoted to a first-class, tracked, documented artefact** (it defines the frame). A city enters the frame if its modal province is Latin-classified, it has a Hanson (2016) urban-population estimate, and it is not Rome — identical to the lodged sample construction, restricted to Latin provinces.

Realised primary frame: 817 cities across 39 Latin provinces. The three frontier classifications flagged in the lodged prereg §9 (Moesia Inferior, Moesia Superior, Sicilia — administrative-Latin but with significant bilingual/Greek populations) remain Latin by the same judgement call, and remain flagged as a known scope choice; a sensitivity excluding the three is available if a reviewer presses (lodged prereg §9 revisit trigger, unchanged).

A5.2 Which analyses are reframed

Analysis	Lodged frame	Amended primary frame	Empire-wide
H3a (f_{within} , primary confirmatory)	empire-wide (“all cities”)	Latin provinces	retained, secondary/context
H3b (deviation-detection on corrected SPAs)	empire + Western subset	Latin provinces primary	empire reported as context
H3c(i) capital contrast; H3c(ii) Moran’s I	empire-wide	Latin provinces	retained, secondary/context
SR1 (OLS log-log scaling comparator)	empire-wide	Latin provinces	retained, secondary/context

H3b’s Crisis-of-the-Third-Century probe already named a “Western-Empire (`province_language == 'Latin'`)” provincial subset in the lodged prereg (§2/§5); this amendment promotes that same Latin frame from a *subset probe within* an empire-wide analysis to the *primary* frame across the cross-sectional tests.

A5.3 Frame-count reconciliation (41 → 39), from evidence

The lodged prereg (§2) states the Latin/Western-Empire subset “covers 41 LIRE provinces (after Rome-exclusion)”. The realised primary frame is **39 provinces**. The difference is fully accounted for and **changes no result**:

- The “41” is the count of Latin-classified provinces in the 2024 notebook’s `province_language_map` (cell 54), Rome excluded.
- Rebuilding the map against LIRE v3.0’s actual `province` field values (`reconcile-province-maps.py`) shows the gap is two provinces that classify Latin but contribute **zero Hanson-matched cities** to the population-scaling frame, plus one spelling normalisation:
 1. **Italia** — LIRE `province` value “Italia” carries **1** inscription; Italian provenances are coded to the eleven Augustan regions (all 11 are in the frame), so “Italia” as a standalone province is redundant and empty of signal. **0** Hanson-matched cities.
 2. **Alpes Graiae** — 77 inscriptions, but **0** carry a Hanson urban-population estimate (0 distinct `urban_context_city` with a population join), so it contributes **0 Hanson-matched cities** to the H3a/SR1/H3c city frame. Its three sibling Alpine provinces (Alpes Cottiae, Alpes Maritimae, Alpes Poeninae) each contribute exactly one city and are in the frame.
 3. **Lugdunensis** → **Lugudunensis** — a 1:1 spelling normalisation to LIRE’s field value (LIRE “Lugdunensis”: 746 inscriptions; “Lugudunensis”: 0). Same province, no count change; documented in the map header.

So **41 Latin-classified provinces — Italia — Alpes Graiae (both 0 cities) = 39 provinces with 1 Hanson-matched city**. The realised frame is 817 cities / 39 provinces whether or not Italia and Alpes Graiae are listed in the map, because neither contributes a city to the population-scaling sample. The reconciliation is a frame-*count* clarification, not a sample change. (Reproduced by `runs/2026-06-06-amendment-02-prep/code/reconcile-province-maps.py` → `outputs/province-reconciliation.csv`.)

A5.4 Realised primary results (Latin frame) — preliminary pending lodgement

Computed and convergence-clean (max R 1.00, min bulk-ESS 2,038, 0 divergences on the Latin fit); reported here for transparency, and confirmatory on lodgement.

Test	Latin frame (817 / 39)	Empire-wide (1,044 / 56), secondary
H3a <code>f_within</code> (unweighted)	0.480, 95% CI [0.401, 0.566] — SUPPORTED ($P > 0.05/0.10/0.20$ all 1.000)	0.299 [0.240, 0.365] — SUPPORTED
<code>_within</code>	0.733 [0.648, 0.820]	0.587 [0.519, 0.657]
SR1 (OLS log-log slope)	0.505 [0.398, 0.611]	0.284
H3c(i) capital contrast (OXREP)	+1.081 [+0.806, +1.408], P=1.000 — SUPPORTED	+0.964 [+0.736, +1.213], SUPPORTED
H3c(ii) Moran’s I (k {5,8,10})	NOT supported (0 of 3 k significant)	not supported

Reading: restricting to the well-covered Latin provinces *strengthens* the within-province population effect and the scaling slope; the Hanson-2021 capital-over-production replication holds on both frames; the residual-clustering half does not replicate on either (the Mundlak province intercepts absorb the broad spatial structure). SR1’s Latin slope (0.505) sits between the empire-frame slope (0.284) and Hanson 2021’s (0.672), and within Carleton et al. (2025)’s 0.3–0.5 elite-subset range — context, not a confirmatory target.

A5.5 Empire-wide results are retained as secondary/context

The empire-wide H3a (`f_within` 0.299) and the empire-wide H3c/SR1 results are **within the original lodged prereg’s “all cities” scope**, are independently verified, and are **retained and reported as secondary/context**

alongside the Latin-frame primary, each with the coverage caveat stated. They are not gated by this amendment and are not withdrawn.

A5.6 Reporting

The paper reports the Latin-frame result as the primary confirmatory finding and the empire-wide result as secondary, with the coverage caveat (“LIRE under-covers Greek-province epigraphy; the empire-wide frame mixes well- and poorly-covered provinces”) stated wherever the empire-wide number appears. No multiplicity correction is introduced between frames: the Latin frame is the primary confirmatory test and the empire-wide frame is descriptive context, not a second confirmatory family.

A6. What does NOT change

- **No model changes.** The H3a Bayesian within-between (Mundlak) negative-binomial regression, its priors, and the `f_within` estimand are exactly as lodged (Decision 12; lodged prereg §3).
- **No decision-rule changes.** The three-way `f_within` rule, the H3c(i) posterior-contrast rule ($P(\text{contrast} > 0) = 0.95$), the H3c(ii) Moran’s-I rule ($I > 0$ at $p < 0.05$ in 2 of {5,8,10}), and SR1’s OLS-log-log comparator are unchanged.
- **No estimand, threshold, prior, or seed changes.**
- **No change to the date-window scope.** H3a/H3c/SR1 operate on date-window-filtered (50 BC – AD 350) counts, as lodged; this amendment does not touch the temporal mixture (that is the separate, forthcoming convention-model amendment, Decision 38).
- **Rome remains excluded**, as lodged.

A7. Changes by preregistration section (a standalone addition, not in-place edits)

Following the project convention (lodged prereg §11 provenance; Amendment 01 §A7), these are recorded as amendment additions, not edits to the lodged text:

- **§2 (Data / subsets).** The Latin-speaking-province subset, lodged as a Western-Empire *subset* for H3b’s Crisis probe (“41 LIRE provinces, Rome excluded”), is **promoted to the primary frame** for H3a/H3b/H3c/SR1. The operational definition is the committed, LIRE-field-aligned province→language map (39 Latin provinces / 817 cities; reconciliation in §A5.3). The map is a first-class tracked artefact.
- **§“Research question” / §3 (H3a).** “Within-province spatial variation in Latin inscription production” is primarily estimated on the **Latin-speaking-province** frame; the empire-wide estimate is reported as secondary/context with the coverage caveat.
- **§3 (H3b, H3c, SR1).** Primary frame = Latin-speaking provinces; empire-wide = secondary/context.
- **§9 (Known limitations).** Adds the dataset-coverage rationale (LIRE under-covers Greek epigraphy) as the stated reason for the Latin-primary frame; the three frontier-province judgement calls (Moesia Inferior/Superior, Sicilia) remain flagged, now as part of the primary frame.

A8. Provenance

- **Decision record:** Decision 36 (`planning/decision-log.md`); cross-sectional sign-off Decision 37 (empire-frame final; Latin-frame amendment-gated).
- **Frame artefact:** `runs/2026-06-04-h3a-confirmatory/data/province-language-map.csv`.
- **Reconciliation:** `runs/2026-06-06-amendment-02-prep/code/reconcile-province-maps.py` → `outputs/province-reconciliation.csv`.
- **Results:** `runs/2026-06-04-h3a-confirmatory/outputs/` — `REPORT.md` (H3a, both frames), `REPORT-latin-h3c-sr1.md`, `REPORT-h3c-i-capital-contrast.md`, `h3a-results.json`, `h3c-latin-results.json`, `sr1-latin-results.json`, `h3c-i-results-oxrep-primary.json`; posteriors `idata-{primary,latin}.nc`.
- **Lodged authority:** the original supplementary (git tag `osf-lodgement-2026-05-20`) remains the lodged authority for all unchanged specifications. This amendment is hosted version-pinned at the project repository and reproduced in the OSF justification field.
- **Repository-state provenance:** the git lodgement tag for this amendment is the reproducibility anchor, distinct from any OSF-assigned amendment identifier.

- **Independence:** separable from Amendment 01 (two-measure framework, lodged 2026-06-04) and from the forthcoming convention-model amendment (Decision 38); the cross-sectional tests reframed here use date-window-filtered counts, not mixture output.